Computational Supplement: Correlation Emergence and the Epps Effect

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Software development v2.1.0 — 2026

Abstract

This supplement records the analytical and simulation companion to "Correlation emergence and the Epps effect in two coupled limit order books" (arXiv:2606.14182). It evaluates the paper's ordinary and fractional correlation build-up factors, reports their sensitivities and identifiability limits, diagnoses finite-grid boundary representation, and verifies a uniform-operational simulation followed by explicit book-specific previous-refresh subordination. The retained numerical evidence includes clock-only and translation-mode coupling-only conformity, a combined no-refit holdout, single-trade and meta-order impact, mid-price-increment and trade-sign autocorrelations, fixed-time order-book shock recovery, and operational/calendar stylised-facts diagnostics. The computations are model-conditional reproducibility evidence, not empirical calibration or mathematical proof.

Contents

1 Scientific and software boundary

The frozen source paper remains the mathematical source of truth. The v2 route adds a diagnosed simulation implementation without changing that source snapshot. The ordinary clock, coupling, and combined curves remain illustrations of the paper's renewal-envelope, exponential-response, and leading-order separability assumptions. The fractional curves evaluate the stated Mittag–Leffler replacement on the non-negative real axis. The reaction-boundary implementation uses the decaying Gaussian convention for which the paper gives a finite half-line moment.

Empirical covariance estimation, market-data calibration, trading use and pointwise kernel equivalence are excluded. The historical Bauer et al. implementation was used during development as a conversion and contrast audit. Its executable code, stored simulations and development-only tests are not shipped in v2. The release retains attribution and the precise mathematical distinctions needed to understand the accepted route.

Both books first evolve on one uniform operational grid. Book-specific calendar clocks act only after the complete operational path exists, by selecting the previous completed state. Calendar waits never enter the density recurrence; no calendar interpolation, extrapolation or nonuniform state update is used.

2 Theory-to-computation map

The ordinary build-up kernel is

$$F(x) = 1 - \frac{1 - e^{-x}}{x}, \qquad x \geq 0, \tag{1}$$

with (F(0)=0) defined by continuity. The implementation uses `expm1` away from the origin and the local series

$$F(x) = \frac{x}{2} - \frac{x^2}{6} + \frac{x^3}{24} - \frac{x^4}{120} + O(x^5) \tag{2}$$

near zero. Its derivative and local rate elasticity are

$$F'(x) = \frac{1 - (1 + x)e^{-x}}{x^2}, \qquad S_F(x) = \frac{xF'(x)}{F(x)}. \tag{3}$$

The normalised ordinary components are

$$C_{\text{clock}}(\Delta) = F(\lambda_{12}\Delta), \tag{4}$$

$$C_{\text{coupling}}(\Delta) = F(\kappa\Delta), \tag{5}$$

$$C_{\text{combined}}(\Delta) = C_{\text{clock}}(\Delta)C_{\text{coupling}}(\Delta). \tag{6}$$

The last expression is the paper's leading-order factorisation; the calculation does not promote it to an exact independence result.

For a characteristic time τ, the fractional component is written

$$F_\alpha(\Delta;\tau) = 1 - E_{\alpha,2}[-(\Delta/\tau)^\alpha], \qquad 0 < \alpha \le 1. \tag{7}$$

The release evaluator uses a deterministic real-axis quadrature for $0 < \alpha < 1$, a small-argument series, and the exact ordinary formula at $\alpha = 1$. Its registered release profiles are $\alpha \in \{0.6, 0.8, 1\}$.

The decaying Gaussian source and its selected half-line moment are

$$q_j(y) = -a_j\mu_j y e^{-\mu_j y^2}, \qquad M_j^+ = \int_0^\infty y q_j(y)\, dy = -\frac{a_j\sqrt{\pi}}{4\sqrt{\mu_j}}. \tag{8}$$

The local frozen-front mapping is

$$\kappa_j = \frac{\gamma_{jk}|M_j^+|}{|\mathcal{L}_j|}, \qquad \kappa = \kappa_1 + \kappa_2. \tag{9}$$

2.1 Source convention and boundary representation

The translated source uses

$$q_j(y) = -a_j\mu_j y \exp(-\mu_j y^2). \tag{10}$$

The retired antecedent implementation used the distinct expression $\exp[-(\mu_j y)^2]$. These expressions coincide only in special parameter cases and are not interchangeable.

For the numerical density state $\phi_j(x, u)$, the production coupling acts through the receiving front's translation mode

$$T_j(x, u) = -\partial_x \phi_j(x, u). \tag{11}$$

A small multiple of T_j translates the resolved zero crossing to first order. The density profile already supplies the reaction-front width, so this implementation does not add a second fixed selector width or an independent boundary layer. The bounded regularised source in the paper remains an analytical weak/local moment closure. Its correspondence with Eq. (11) is asserted after projection onto front displacement, not as pointwise identity between kernels.

This clarification is the development link to the Bauer antecedent. The conversion audit identified the nonuniform update, the different Gaussian convention and the interface ambiguity. The accepted Angstmann–Gebbie route then separated uniform operational dynamics from calendar observation and tested the translation-mode implementation first as a component and then in a combined no-refit holdout. The retired implementation remains recoverable in the sealed pre-v1.9.3 gates and repository history, but is not an executable dependency of this release.

3 Ordinary Epps components and sensitivity

Figure 1 is the analytical baseline used to interpret the simulation results. The three panels nondimensionalise the clock rate to one and vary the response-to-clock rate ratio. The curves separate the two mechanisms before forming the combined product.

The short-scale limit of each component is linear in its dimensionless argument, so its rate elasticity approaches one. Once the component reaches its plateau, the elasticity approaches zero. For the product,

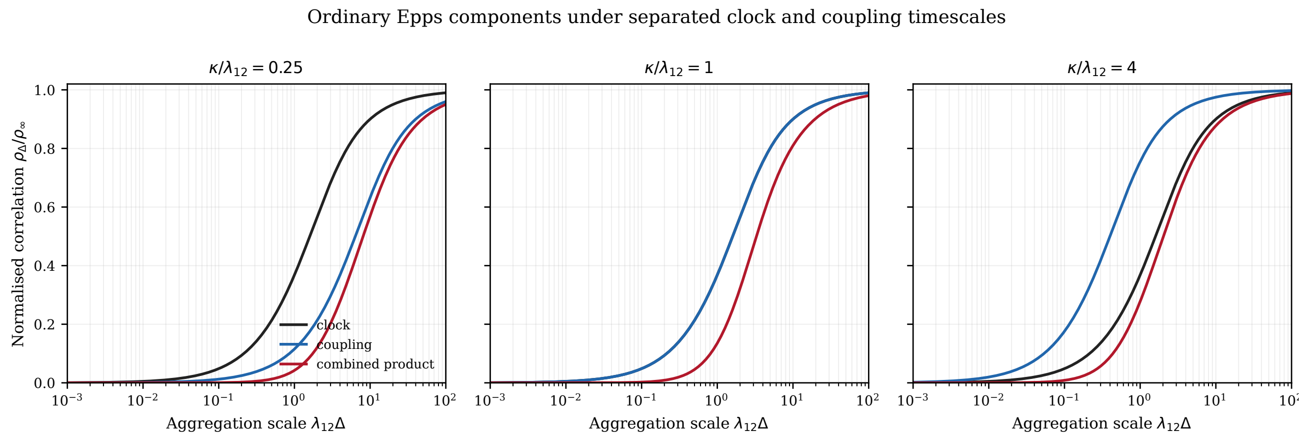

Figure 1: **Ordinary Epps components under separated clock and coupling timescales.** The clock curve is $F(\lambda_{12}\Delta)$, the coupling-response curve is $F(\kappa\Delta)$, and the combined curve is their separable product. The horizontal axis is aggregation scale in clock characteristic-time units and the vertical axis is ρ_Δ/ρ_∞. Panels use $\kappa/\lambda_{12} = 0.25, 1, 4$, with $\lambda_{12} = 1$ and $\rho_\infty = 1$. The figure shows how the mechanisms contribute across separated and comparable timescales and supplies the analytic overlay baseline for v2.0.0. It supports the component decomposition under the paper's renewal-envelope and separability assumptions; it does not fit market data, simulate an order book, prove statistical independence, or establish the approximation outside those assumptions.

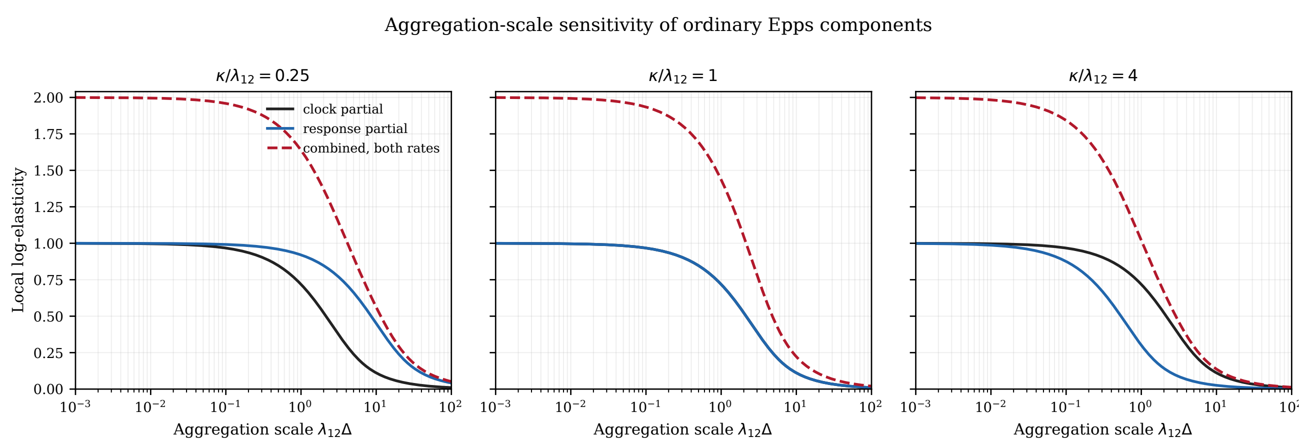

Figure 2: **Aggregation-scale sensitivity of ordinary Epps components.** Curves report $S_F(x) = xF'(x)/F(x)$ for clock refresh and coupling response. The dashed red curve is the log-sensitivity of the combined product when both rates receive the same proportional perturbation. Panels use the Figure 1 rate ratios. Sensitivity tends to one per component in the resolved short-scale limit and to zero on the long-scale plateau, locating the aggregation range that contains rate information. The figure supports local, model-conditional sensitivity statements; it is not a global-identifiability proof, an empirical uncertainty estimate, or evidence that either rate can be recovered without adequate short-scale observations.

the partial log-sensitivity to each rate is inherited from that component, while a common proportional perturbation of both rates has the sum of the two partial elasticities.

The combined ordinary curve is invariant under exchange of the clock and response rates. Moreover,

$$C_{\text{combined}}(\Delta) = \frac{\lambda_{12}\kappa}{4}\Delta^2 + O(\Delta^3) \tag{12}$$

at short scale. Thus the leading combined curve alone reveals the rate product rather than assigning each timescale to its mechanism. The simulation results below therefore retain independently declared refresh and response rates.

4 Fractional order sensitivity

From Eq. (7),

$$F_\alpha(\Delta;\tau) = \frac{(\Delta/\tau)^\alpha}{\Gamma(\alpha+2)} + O(\Delta^{2\alpha}). \tag{13}$$

A combined fractional curve therefore has leading power $\Delta^{\alpha_c+\alpha_r}$. Equal-sum order pairs are deliberately included in Figure 3 to show the resulting local confounding.

5 Conditional reaction-boundary propagation

Equation (9) yields conditional log-elasticities $+1, +1, -1/2, -1$ with respect to coupling strength, source amplitude, source width, and front-slope magnitude. Figure 4 changes one quantity at a time. This is a propagation experiment: multiple structural parameter combinations generate the same effective response rate and hence the same analytic Epps curve.

Three distinct notions must not be collapsed into a single "boundary thickness": the local reaction-front slope $|\mathcal{L}_j|$, the continuum selector regularisation ε, and the lattice spacing Δx. The first enters the structural response mapping. The second regularises side selection. The third is a numerical resolution parameter.

6 Continuum selector and finite-grid representation

The paper's smooth selector is

$$W(y,z;\varepsilon) = \frac{1}{2}\left[1 + \tanh\left(\frac{yz}{\varepsilon}\right)\right]. \tag{14}$$

For a centred symmetric full domain, $yq_j(y)$ is even and the hyperbolic-tangent contribution is odd. Consequently the selected first moment is independent of ε in the continuum calculation. Finite domains, misaligned numerical selectors, and coarse grids can break this cancellation. Figure 5 diagnoses those representation effects and propagates them through an effective response rate.

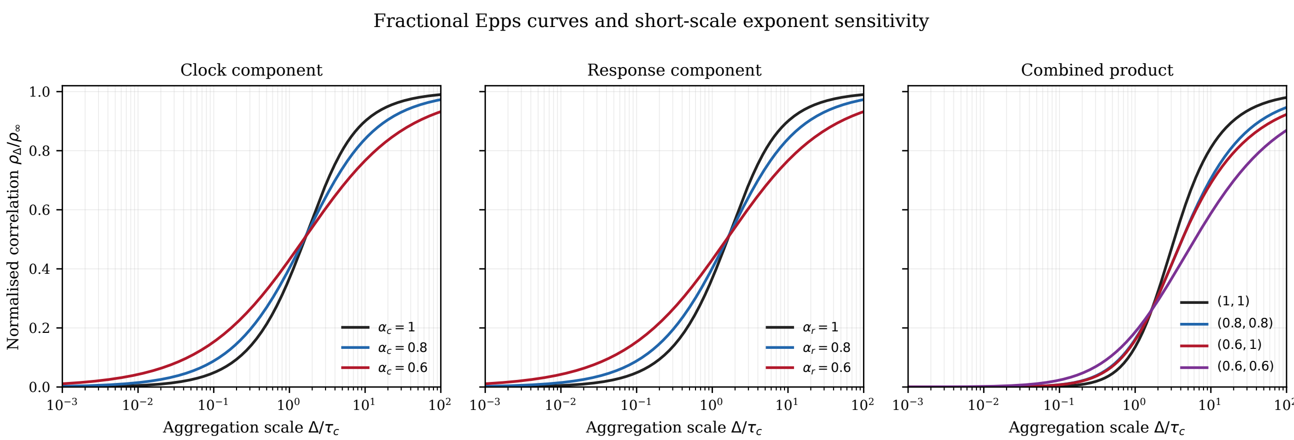

Figure 3: **Fractional Epps curves and short-scale exponent sensitivity.** Clock and response components use Eq. (7), with unit characteristic times and the diagnosed evaluator recorded in the numerical-design note; $\alpha = 1$ recovers the ordinary kernel. Combined curves use order pairs $(1, 1), (0.8, 0.8), (0.6, 1), (0.6, 0.6)$. The equal-sum cases $(0.8, 0.8)$ and $(0.6, 1)$ share leading short-scale power $\Delta^{1.6}$, exposing local confounding even when later-scale curvature differs. The figure shows formula sensitivity to fractional order and the ordinary limit. It does not identify both orders from a combined curve alone, attach an unconditional "slower" interpretation across dimensional conventions, or validate a fractional market-clock mechanism empirically.

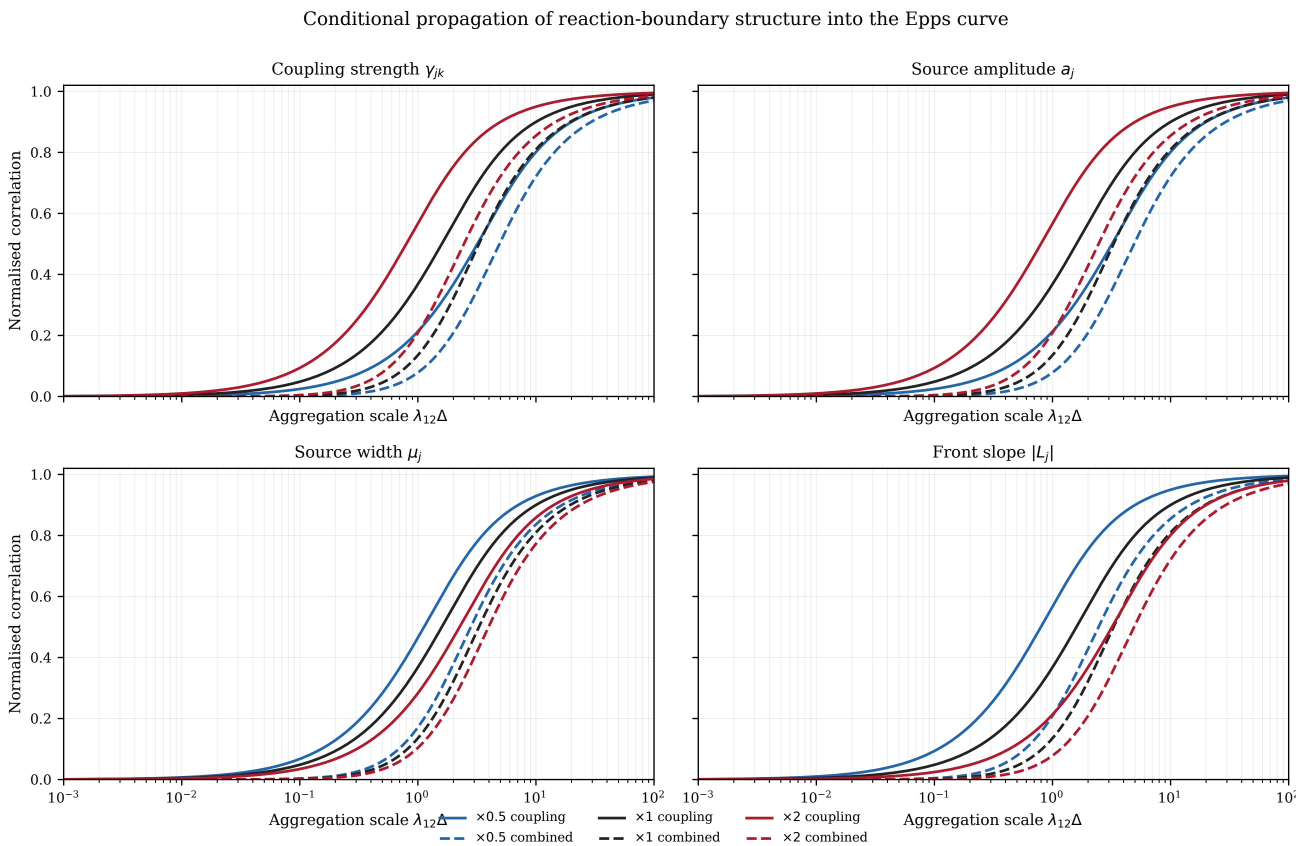

Figure 4: **Conditional propagation of reaction-boundary structure into the Epps curve.** The response rate is varied through Eqs. (8)–(9). Two symmetric books, unit source amplitude, width and front-slope magnitude, and $\gamma_{jk} = 2/\sqrt{\pi}$ give baseline total response rate $\kappa = 1$. Each panel changes one input by factors $0.5, 1, 2$, holding the others and the unit clock rate fixed; solid curves are coupling factors and dashed curves their combined products with the clock factor. This supports conditional propagation through the analytic response mapping. It does not uniquely infer reaction-front thickness, source width, or coupling strength from an Epps curve and does not claim pointwise equivalence to the legacy lattice coupling.

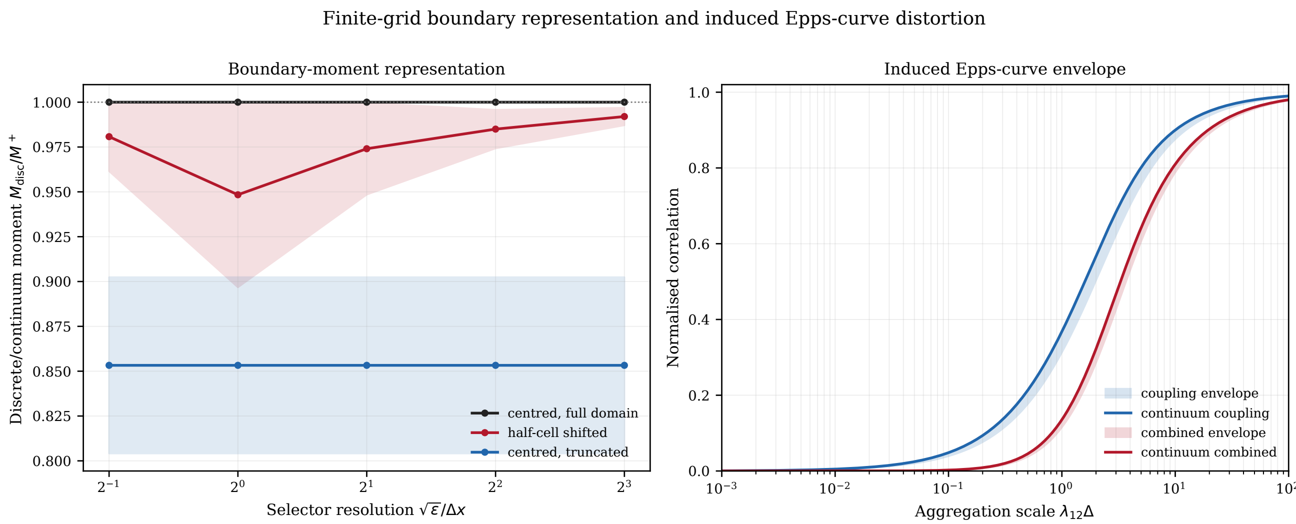

Figure 5: **Finite-grid boundary representation and induced Epps-curve distortion.** The left panel reports discrete-to-continuum first-moment ratios as selector resolution $\sqrt{\varepsilon}/\Delta x$, lattice spacing, selector alignment, and domain margin vary. Lines give the midpoint and bands the range over four lattice spacings. The centred full-domain case is the parity control; half-cell selector displacement and truncation measure numerical representation errors. The right panel propagates all diagnosed effective-rate ratios through the coupling and combined Epps formulas; lines are continuum baselines and bands are discrete-representation envelopes. Structural parameters are fixed. The figure supports convergence and numerical-sensitivity statements and defines resolution metadata required for v2 overlays. It does not introduce a new continuum Epps mechanism or treat selector width, lattice spacing, and reaction-front thickness as interchangeable.

7 Calendar-time representation and memory diagnostic

The aggregation variable in Figures 1–5 remains the observation-window lag, but is reported relative to the clock characteristic time. Writing

$$x = \lambda_{12}\Delta t = \frac{\Delta t}{\tau_c}, \qquad \tau_c = \lambda_{12}^{-1}, \tag{15}$$

shows that a calendar-time presentation requires only a declared time convention. For an ordinary exponential memory with rate r,

$$h_r(u) = e^{-ru}, \qquad F(r\Delta t) = 1 - \frac{1}{\Delta t}\int_0^{\Delta t} h_r(u)\, du. \tag{16}$$

For the clock mechanism, $h_{\lambda_{12}}$ is the no-refresh survival. For the coupling mechanism, h_κ is the relaxation survival and equals the normalised positive-lag exponential covariance kernel used in the paper. Figure 6 therefore presents the same analytical factors as Figure 1, not a new Epps mechanism.

The distinction from the simulation output is substantive. A sample autocorrelation requires a generated path, a declared estimator and finite-sample uncertainty. The deterministic inset exposes only the memory functions that enter Eq. (16). Figure 11 reports the separate path-derived log-mid increment and trade-sign autocorrelations without retrospectively relabelling this analytical diagnostic.

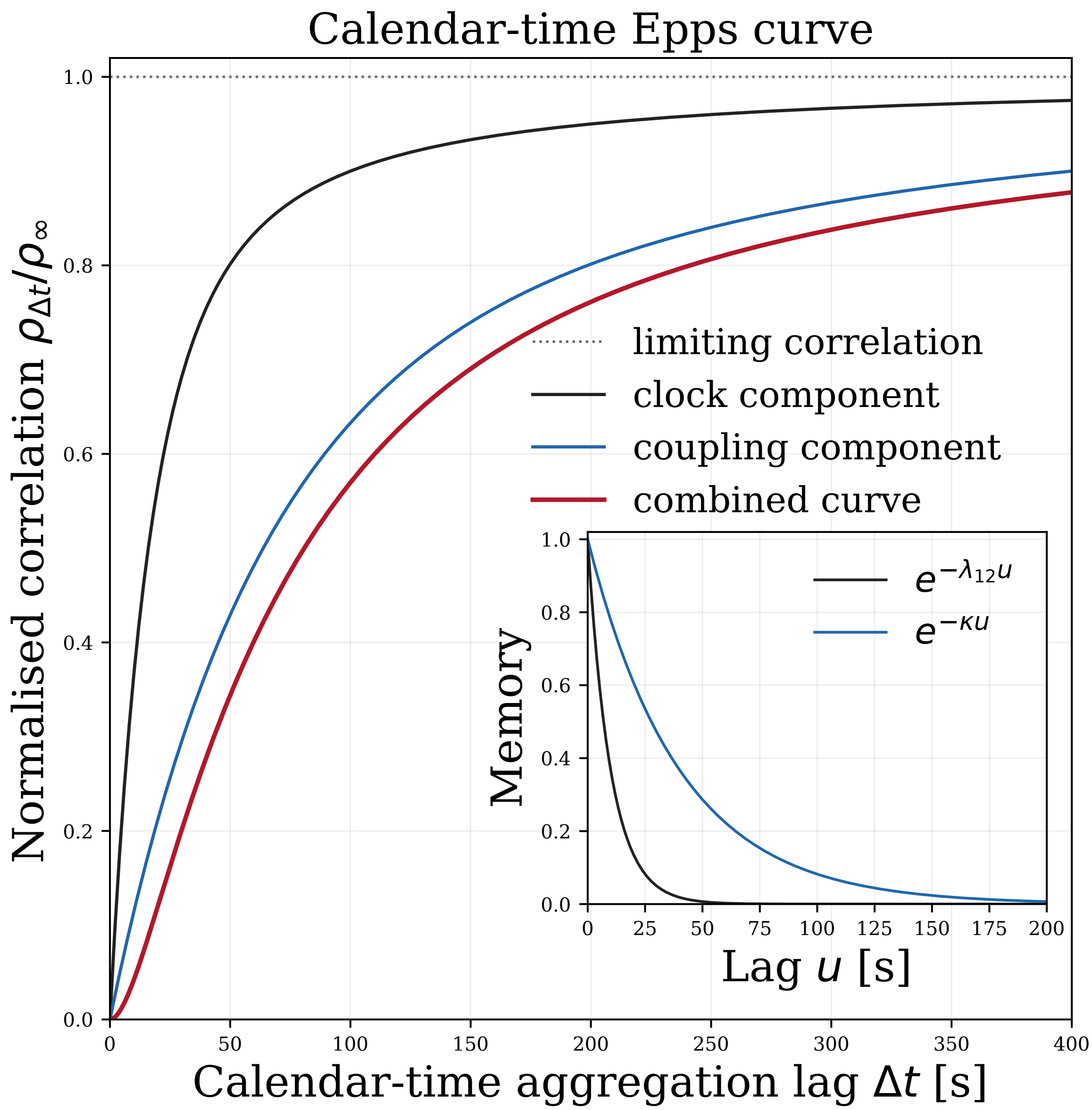

Figure 6: **Calendar-time Epps curve.** The square main panel rewrites the Figure 1 slow-response profile on a linear calendar-time axis using the explicitly illustrative convention $\lambda_{12} = 0.1\,\mathrm{s}^{-1}$, $\kappa = 0.025\,\mathrm{s}^{-1}$, and $\rho_\infty = 1$, corresponding to $\tau_c = 10$ s and $\tau_r = 40$ s. Black, blue and red curves are the clock factor, coupling factor and their leading-order separable product; the grey dotted line marks the normalised limiting correlation $\rho_{\Delta t}/\rho_\infty = 1$. The enlarged inset reports $h_c(u) = e^{-\lambda_{12}u}$ and $h_r(u) = e^{-\kappa u}$: respectively the no-refresh survival and coupling-relaxation survival, with the latter also equal to the normalised positive-lag exponential covariance kernel. The seconds scale is not fitted to data, and the inset is neither a path autocorrelation nor a power-spectrum estimate. Path-derived autocorrelations are reported separately in Figure 11.

8 Uniform-operational simulation

Let $u_n = n\Delta u$ be the common operational grid. Both density fields are advanced on this grid before any calendar observation is constructed. External innovations are book-specific, the translation-mode coupling uses Eq. (11), and labelled order events modify the density state at declared operational steps. The boundary price is the accepted unique interior zero crossing of the signed density field.

For book j, let $\tau_{j,m}$ be its completed calendar refresh events. The software first selects the latest event at or before t, then the latest uniform operational node at or before that event:

$$m_j(t) = \max\{m : \tau_{j,m} \le t\}, \qquad n_j(t) = \max\{n : u_n \le \tau_{j,m_j(t)}\}, \qquad P_j^{\mathrm{cal}}(t) = P_j^{\mathrm{op}}(u_{n_j(t)}). \tag{17}$$

Equation (17) is previous-refresh subordination. It selects an already completed state and never changes an operational update interval.

8.1 Executable numerical algorithms

The two algorithms below state the implemented separation between operational evolution and calendar observation. They are deliberately more explicit than the schematic model equations: all coupling fields are formed from one frozen pre-step state, and all calendar observations select states from an already completed operational path. The notation follows the code rather than introducing a second numerical scheme.

Algorithm 1. Coupled operational-time reaction-boundary evolution

Required: a uniform price grid x_i, operational step Δu, density histories $\{\varphi_m^{i(j)}\}_{m\le n}$, registered boundaries $p_n^{(j)}$, source fields $s_n^{i(j)}$, external event fields $\delta_n^{i(j)}$, and ordered rates $\kappa_{jk} \ge 0$.

Returned: the next densities and boundaries, every directed coupling field, the net source field, and the term-level diagnostics.

1. Freeze the current densities and boundaries before either book is updated:
$$\varphi_{n,\mathrm{pre}}^{i(j)} = \varphi_n^{i(j)}, \qquad p_{n,\mathrm{pre}}^{(j)} = p_n^{(j)}. \tag{18}$$

2. For every ordered pair $j \leftarrow k$, compute the spread and the receiving-front translation mode from that same snapshot:
$$z_{jk,n} = p_{n,\mathrm{pre}}^{(j)} - p_{n,\mathrm{pre}}^{(k)}, \qquad v_{j,n}^{\Delta x}(x_i) = -\partial_x^{\Delta x}\varphi_{n,\mathrm{pre}}^{i(j)}. \tag{19}$$
The implementation uses the second-order finite-difference gradient and sets the two outer values of $v_{j,n}^{\Delta x}$ to zero.

3. Form and sum the directed coupling fields received by book j:
$$\ell_{T,n}^{i(j,k)} = -\kappa_{jk} z_{jk,n} v_{j,n}^{\Delta x}(x_i), \qquad \ell_{T,n}^{i(j)} = \sum_{k\ne j} \ell_{T,n}^{i(j,k)}. \tag{20}$$
Thus either $\kappa_{jk} = 0$ or $z_{jk,n} = 0$ makes the corresponding directed field exactly zero.

4. Assemble the field supplied to the source-input channel of the fixed operational recurrence,
$$c_n^{i(j)} = s_n^{i(j)} + \delta_n^{i(j)} + \ell_{T,n}^{i(j)}, \tag{21}$$
while retaining its three components separately in the diagnostic record.

5. Advance both books from the frozen histories:

$$\varphi^{i(j)}_{n+1,\mathrm{raw}} = \mathcal{U}^{(j)}_{\Delta u}\Big[\{\varphi^{(j)}_m\}_{m\le n}, c^{(j)}_n\Big]. \tag{22}$$

The operator $\mathcal{U}^{(j)}_{\Delta u}$ is the implemented fixed-step DTRW recurrence and owns the history, survivor, transport, cancellation and source increments. The additive field $c^{(j)}_n$ is its source input, not a separate state update.

6. Impose the declared zero Dirichlet condition on the raw field:

$$\varphi^{0(j)}_{n+1} = \varphi^{I(j)}_{n+1} = 0. \tag{23}$$

7. Form the admissible exact-grid zeros and strict adjacent sign crossings,

$$\mathcal{Z}^{(j)}_{n+1,\mathrm{exact}} = \Big\{x_i : 1 \le i \le I-1,\ \varphi^{i(j)}_{n+1} = 0,\ \varphi^{i-1(j)}_{n+1}\varphi^{i+1(j)}_{n+1} < 0,\ |L^{(j)}_i| \ge L_{\min}\Big\}, \tag{24}$$

$$\mathcal{Z}^{(j)}_{n+1,\mathrm{cross}} = \left\{x_i - \frac{\varphi^{i(j)}_{n+1}}{L^{(j)}_{i+1/2}} :\ \varphi^{i(j)}_{n+1}\varphi^{i+1(j)}_{n+1} < 0,\ |L^{(j)}_{i+1/2}| \ge L_{\min}\right\}, \tag{25}$$

where

$$L^{(j)}_i = \frac{\varphi^{i+1(j)}_{n+1} - \varphi^{i-1(j)}_{n+1}}{2\Delta x}, \qquad L^{(j)}_{i+1/2} = \frac{\varphi^{i+1(j)}_{n+1} - \varphi^{i(j)}_{n+1}}{\Delta x}. \tag{26}$$

8. Set $\mathcal{Z}^{(j)}_{n+1} = \mathcal{Z}^{(j)}_{n+1,\mathrm{exact}} \cup \mathcal{Z}^{(j)}_{n+1,\mathrm{cross}}$ and require it to be non-empty. Unique selection requires exactly one candidate. Nearest-previous selection requires a unique minimum distance from $p^{(j)}_n$ and rejects distance ties.

9. Return

$$p^{(j)}_{n+1} = \mathcal{B}_j\Big[\mathcal{Z}^{(j)}_{n+1}; p^{(j)}_n\Big] \tag{27}$$

together with the complete candidate count, selected slope, distance to the domain edge, fields, sources and recurrence contributions.

Algorithm 2. Book-specific previous-refresh calendar observation

Required: completed operational prices $p^{(j)}(u_n)$ on the uniform grid, finite strictly increasing refresh-event times $0 = \tau_{j,0} < \tau_{j,1} < \cdots < \tau_{j,M_j}$ for each book, a finite nondecreasing query grid t_r, and integer lags h_q.

Returned: subordinated prices, selected refresh events and operational indices, return-component sums, optional exact conditional-reference components, and support records.

1. Restrict the query grid to the common realised support,

$$0 \le t_0 \le \cdots \le t_R \le \min\{u_N, \tau_{1,M_1}, \tau_{2,M_2}\}. \tag{28}$$

No extrapolation or terminal-state clamping is permitted.

2. At each query time select first the previous refresh event and then the previous uniform operational state:

$$m_j(t_r) = \max\{m : \tau_{j,m} \le t_r\}, \qquad n_j(t_r) = \max\{n : u_n \le \tau_{j,m_j(t_r)}\}. \tag{29}$$

This nested map is the exact software realisation of the previous-completed- state inverse; it performs no interpolation.

3. Construct the calendar observation

$$P_{\Delta}^{(j)}(t_r) = p^{(j)}\left(u_{n_j(t_r)}\right). \tag{30}$$

4. For every lag h_q, form returns and their pathwise sufficient statistics:

$$R_{q,r}^{(j)} = P_{\Delta}^{(j)}(t_{r+h_q}) - P_{\Delta}^{(j)}(t_r), \tag{31}$$

$$S_q^{12} = \sum_r R_{q,r}^{(1)} R_{q,r}^{(2)}, \qquad S_q^{jj} = \sum_r (R_{q,r}^{(j)})^2, \tag{32}$$

$$\widehat{\rho}_q = \frac{S_q^{12}}{\sqrt{S_q^{11} S_q^{22}}}. \tag{33}$$

5. Record the selected operational intervals $J_{j,q,r} = (a_j, b_j]$, where

$$(a_1, b_1, a_2, b_2) = \Delta u\left(n_1(t_r), n_1(t_{r+h_q}), n_2(t_r), n_2(t_{r+h_q})\right), \tag{34}$$

and their realised overlap

$$\Theta_{q,r} = \max\{0, \min(b_1, b_2) - \max(a_1, a_2)\}. \tag{35}$$

6. When the symmetric reduced conditional reference is requested, use $\kappa = \kappa_{12} + \kappa_{21} > 0$ and evaluate

$$K_{q,r} = e^{-\kappa|b_1-b_2|} - e^{-\kappa|b_1-a_2|} - e^{-\kappa|a_1-b_2|} + e^{-\kappa|a_1-a_2|}, \tag{36}$$

$$C_{q,r}^{12} = \Theta_{q,r} - \frac{K_{q,r}}{2\kappa}, \tag{37}$$

$$V_{q,r}^{(j)} = (b_j - a_j) - \frac{e^{-\kappa(b_j-a_j)} - 1}{\kappa}. \tag{38}$$

The minus sign before the exponential increment in Eq. (38) is intentional: because $e^{-\kappa d} - 1 \leq 0$, the Ornstein–Uhlenbeck spread contribution adds to each marginal variance and subtracts from the cross-covariance. Under identity clocks these components recover the implemented coupling-only covariance and correlation exactly to the registered numerical tolerance.

7. Pool component triples across the G independent path groups before taking a ratio. For uncertainty, form each delete-one-group ratio

$$\widehat{\rho}_q^{(-g)} = \frac{S_q^{12} - S_{gq}^{12}}{\sqrt{(S_q^{11} - S_{gq}^{11})(S_q^{22} - S_{gq}^{22})}} \tag{39}$$

and the path-group jackknife standard error

$$\mathrm{se}_{\mathrm{JK}}(\widehat{\rho}_q) = \left[\frac{G-1}{G}\sum_{g=1}^{G}\left(\widehat{\rho}_q^{(-g)} - \overline{\rho}_q^{(-)}\right)^2\right]^{1/2}. \tag{40}$$

The first algorithm owns the density dynamics and boundary extraction. The second owns only observation and estimator construction. Identity clocks recover the stored operational prices at matching nodes; distinct clocks alter only the calendar image of the completed paths.

8.2 Final theory-to-simulation Epps comparison

Figure 7 is the primary release result. The component inputs, clock paths, holdout structure and parameter values were fixed before the combined comparison. The leading-order product remains the analytical paper envelope. The same-clock conditional reference is the exact finite-step conditional moment evaluated at the operational indices selected by the same realised clocks as the simulation; it is not a fitted replacement.

8.3 Translation-mode component

The translation-mode coupling is tested separately under identity clocks before it is used in the combined route. Deterministic signed-front relaxations diagnose the registered response rate; stochastic holdout paths test normalized covariance and the distinct finite-scale return correlation. The covariance scale is frozen from a disjoint calibration ensemble.

9 Event impact and dependence diagnostics

Market-order events consume opposing density at the active boundary and are recorded on an immutable event tape. Impact is measured by pairing each event path with an otherwise identical zero-input control. Responses are aggressor-signed log-boundary differences, so own and cross effects use one common linear scale. Calendar versions are obtained only after the operational paths are complete, using Eq. (17).

A meta-order is a declared schedule of child market orders with fixed total quantity and book. Equal-volume horizon comparisons distinguish schedule timing from a true participation rate, which would require background market-order volume not present in this model.

For a sequence X_n, let $\bar{X}_{L,k}$ and $\bar{X}_{R,k}$ denote the separate sample means of $\{X_n\}_{n=1}^{N-k}$ and $\{X_{n+k}\}_{n=1}^{N-k}$. The registered positive-lag statistic is the Pearson correlation of those two overlapping slices,

$$\widehat{\rho}_X(k) = \frac{\sum_{n=1}^{N-k}(X_n - \bar{X}_{L,k})(X_{n+k} - \bar{X}_{R,k})}{\sqrt{\left[\sum_{n=1}^{N-k}(X_n - \bar{X}_{L,k})^2\right]\left[\sum_{n=1}^{N-k}(X_{n+k} - \bar{X}_{R,k})^2\right]}}, \qquad \widehat{\rho}_X(0) = 1. \tag{41}$$

The price statistic is applied to log-mid increments, never to price levels. Trade-sign diagnostics distinguish ground-truth aggressor signs, a quote/midpoint rule and the historically inherited tick rule. Event-time signs and calendar-binned signed flow are separate estimands.

9.1 Fixed-time order-book shock recovery

Figure 12 follows one buy market order in book 1 through the current fixed-grid operational dynamics. Immediately before the declared operational step, the order consumes opposing density from the reaction boundary outwards. If M_{1,n_*} denotes the resulting density increment, then

$$\varphi_{1,n_*}^{+} = \varphi_{1,n_*}^{-} + M_{1,n_*}, \qquad \Delta x \sum_i |M_{1,n_*,i}| = Q. \tag{42}$$

The event is applied once and only to book 1. The second book remains active through the receiving-front translation-mode coupling, although its density and the coupling contribution are not separately drawn.

For a completed operational step the stored density satisfies

$$\varphi_{j,n+1} = H_{j,n} + \varphi_{j,n}^{+} + C_{j,n} + A_{j,n} + L_{j,n} + B_{j,n}, \tag{43}$$

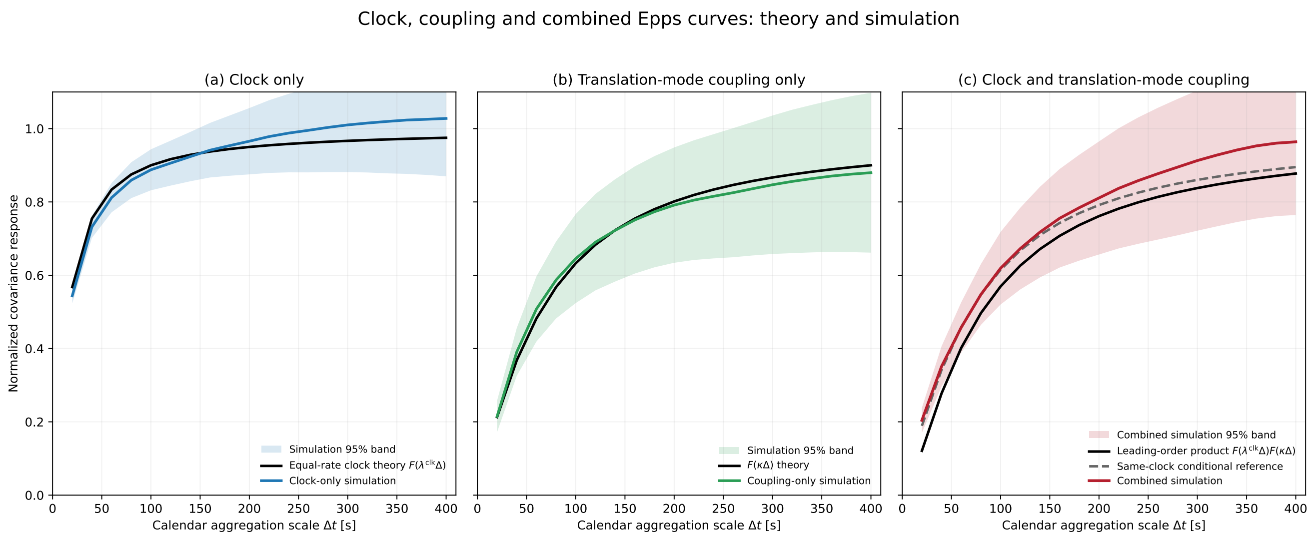

Figure 7: **Clock, coupling and combined Epps curves: theory and simulation.** The three square panels share one linear 0–400 second and 0–1.1 normalized-covariance scale. Panel (a) compares the clock-only simulation with the exact equal-rate previous-refresh factor $F(\lambda^{\mathrm{clk}}\Delta)$, where each book has $\lambda^{\mathrm{clk}} = 0.1\,\mathrm{s}^{-1}$ rather than the $0.2\,\mathrm{s}^{-1}$ pooled minimum-wait rate. Panel (b) compares the translation-mode coupling simulation with $F(\kappa\Delta)$, using $\kappa = 0.025\,\mathrm{s}^{-1}$. Panel (c) compares the combined no-refit holdout with the paper's leading-order separable product $F(\lambda^{\mathrm{clk}}\Delta)F(\kappa\Delta)$ and the same-clock conditional reference. The latter evaluates the reduced finite-step conditional moment at the same realised previous-refresh indices as the simulation; it is not a fitted curve. Its combined RMSE is 0.039719, standardized RMSE is 0.455132, and pointwise normal-band coverage is complete. No parameter, normalization or time map is refitted.

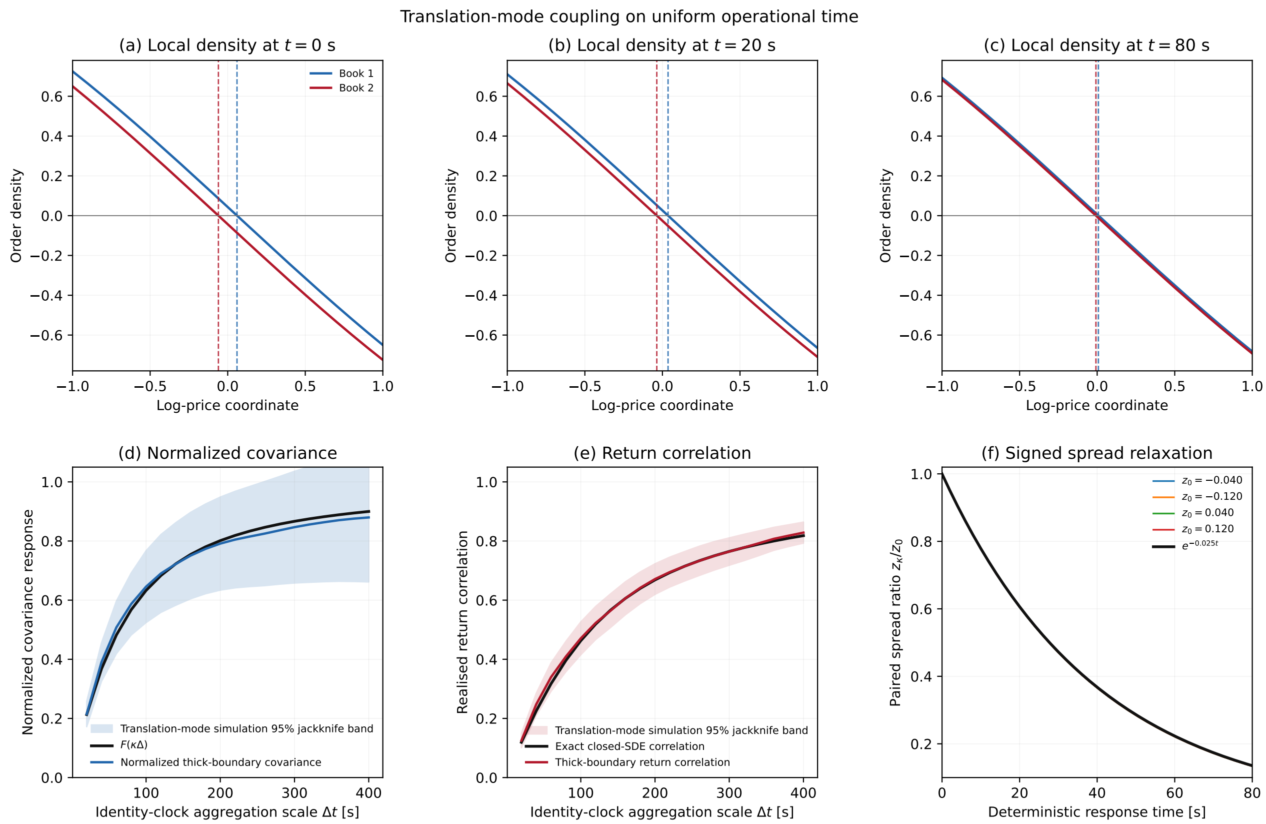

Figure 8: **Translation-mode coupling on uniform operational time.** Holdout thick-boundary paths are compared with the analytical normalized covariance response and the distinct finite-scale return correlation. Lower panels verify signed deterministic front relaxation and residuals at the frozen response rate. No clock, interpolation or parameter fit enters this component gate.

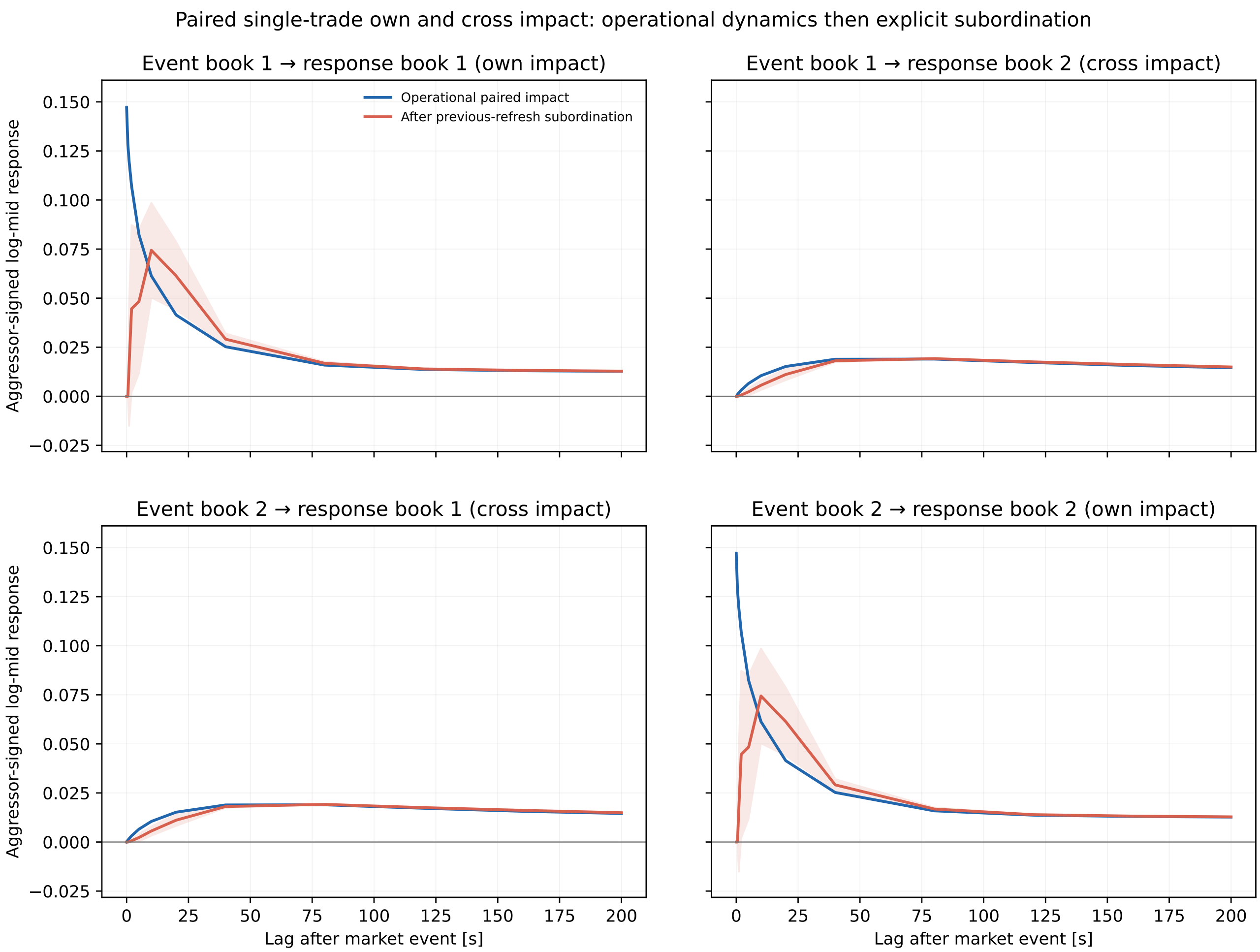

Figure 9: **Paired single-trade own and cross impact.** One labelled market order is compared with a matched control in operational and previous-refresh calendar views. The figure reports own and cross boundary responses on a common linear scale. It is a declared model experiment, not an empirical trade calibration.

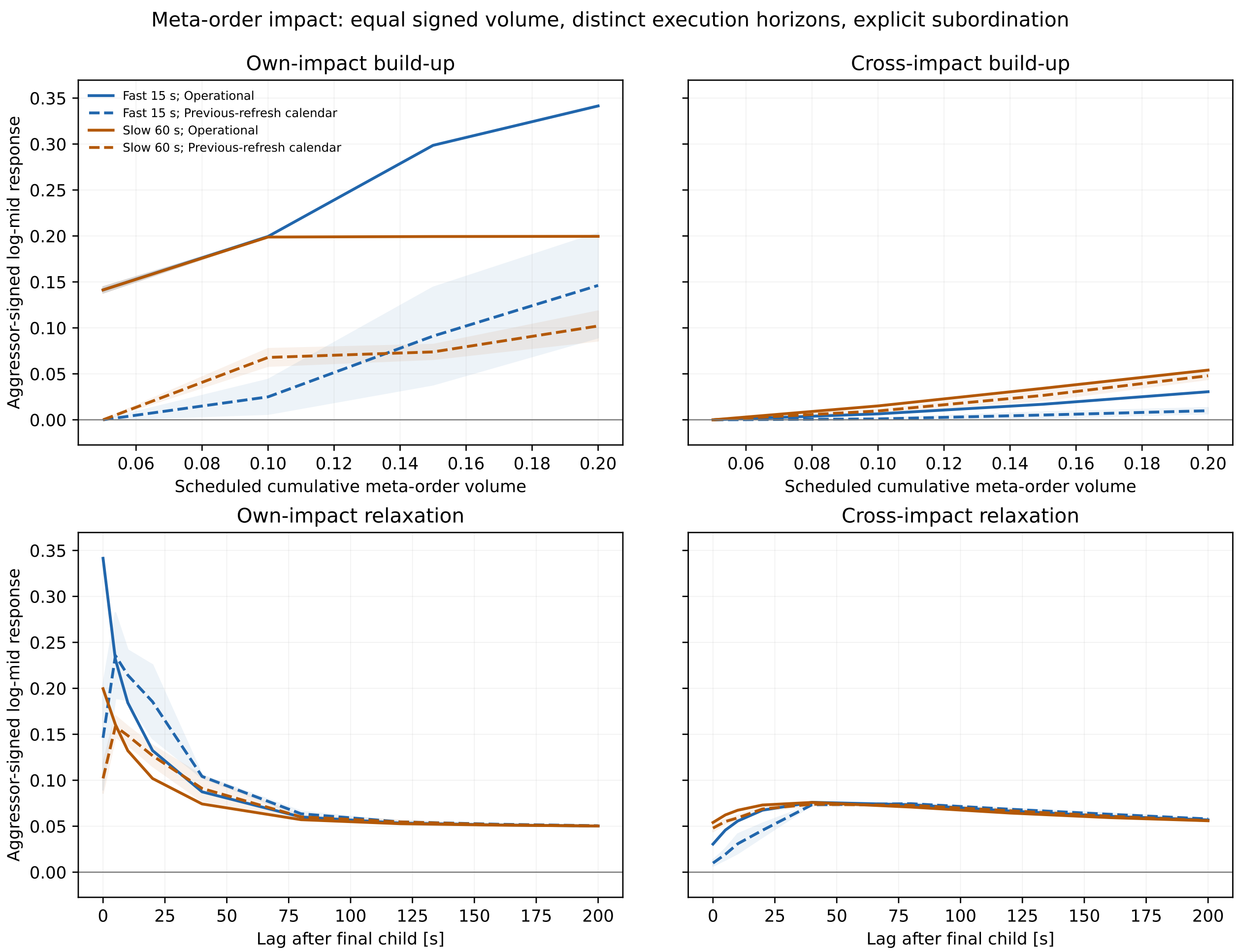

Figure 10: **Scheduled meta-order own and cross impact.** Matched schedules report trajectory, peak, relaxation and equal-volume horizon comparisons under operational and explicitly subordinated calendar observations. Schedule timing, cross-impact catch-up and long-lag convergence are kept as distinct observables.

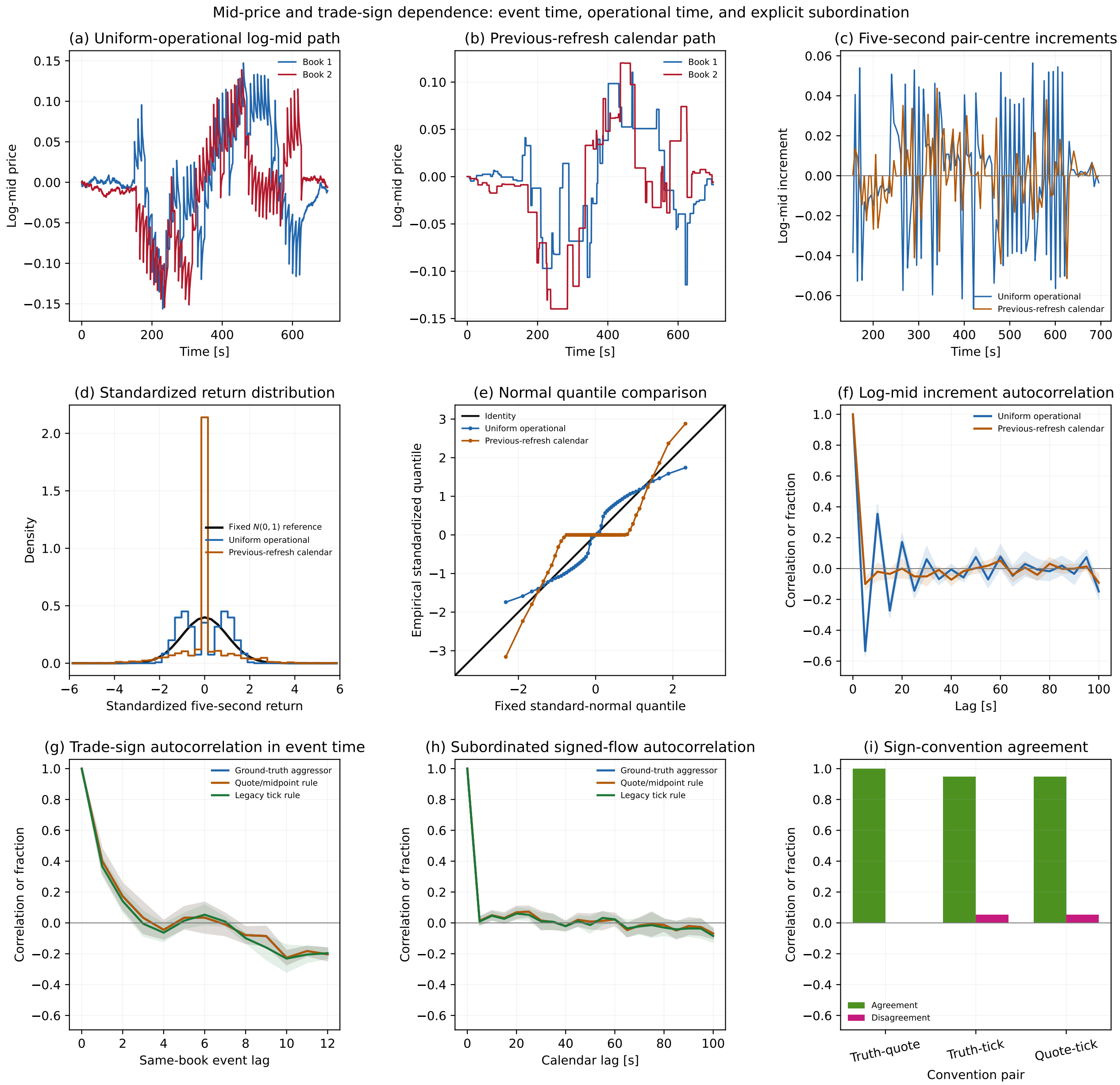

Figure 11: **Mid-price and trade-sign autocorrelations.** Panels report log-mid increment autocorrelation for uniform operational and previous-refresh calendar observations, same-book event-time trade-sign autocorrelation for three sign conventions, subordinated signed-flow autocorrelation, and sign-convention agreement. Price-level autocorrelation is excluded. The finite Markov persistence is an estimator fixture, not empirical calibration or an endogenous long-memory claim. The operational five-second ACF is a registered finite periodic-schedule diagnostic, and its detailed curve is phase-sensitive to that schedule.

where H is the transport or memory contribution,

$$A_{j,n} = \Delta u\, q_j, \qquad C_{j,n} = \left(e^{-\nu_j \Delta u} - 1\right) \varphi^{+}_{j,n}, \qquad L_{j,n} = \Delta u\, \ell_{j\leftarrow k}, \tag{44}$$

and B is the imposed outer-boundary correction. The plotted arrivals, removals and market-order impulse are density contributions rather than rates. All are drawn on the same density scale. In particular, the green curve is $A_{j,n} = \Delta u\, q_j$ and is small relative to φ_j because $\Delta u = 0.005$; it is not independently rescaled. The unplotted terms are retained in the machine-readable ledger.

Two differences from Bauer et al. Figure 2 follow from the present model. The accepted configuration has $\nu_1 = \nu_2 = 0$, so $C_{j,n} = 0$ and the gold removal curve is identically zero. It is retained to preserve the plotted object map; no positive cancellation rate is introduced for visual agreement. Second, boundary-outward consumption produces a short zero-density interval in the intermediate 0^+ state. That state has no unique simple zero crossing. The marker in panel (b) is therefore the last registered boundary p_1^-; the first completed fixed-grid step restores a unique registered reaction boundary. Later markers are independently recovered from the corresponding density states.

The first completed step gives an own-boundary displacement of 0.0997 in log-price units. The displacement reaches 0.111 at 2 seconds and then decreases to 0.0160 at 80 seconds. Thus the computed late state is substantially relaxed but remains displaced from the matched control over the displayed horizon.

The calculation uses fixed $\Delta x = 0.1$ and $\Delta u = 0.005$ model-time units, with 100 seconds per model-time unit. Innovations are set to zero so that the chronology isolates event consumption, diffusion, source renewal and coupling. A matched no-event control remains in the numerical checks but is not added to the nine-panel figure. The result recovers the structure and scientific question of the earlier figure; it does not assert numerical identity with its non-uniform-time trajectory or parameterisation.

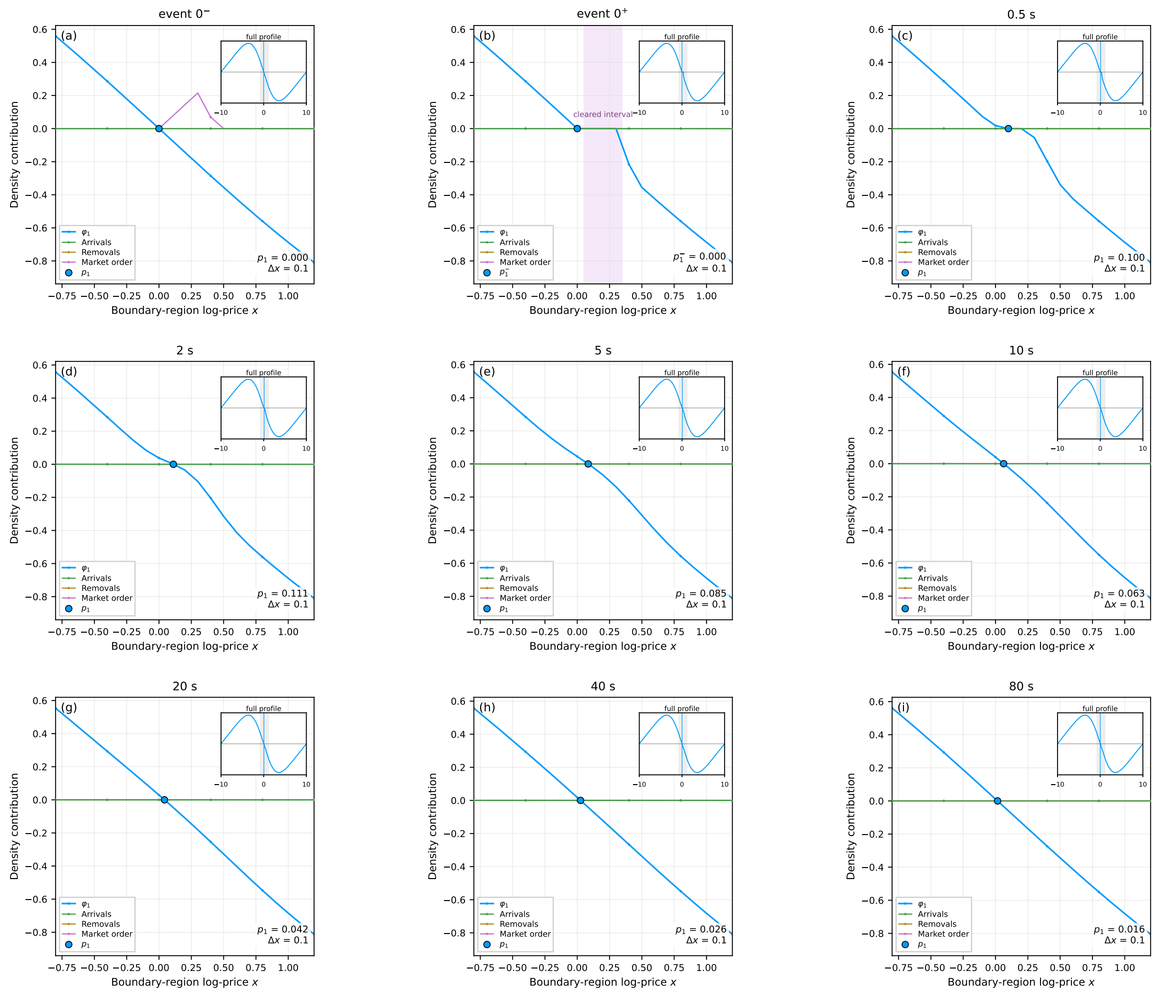

Figure 12: **Fixed-time order-book shock recovery.** Nine snapshots follow a buy market order in book 1 from the pre-event state 0^-, through the post-consumption state 0^+, to 80 seconds of fixed operational-time evolution. Blue is the signed density φ_1, green is the per-step arrival contribution, gold is the per-step removal contribution, magenta is the market-order density increment shown in panel (a), and the blue marker is the registered reaction boundary. The order consumes opposing liquidity from the boundary outwards. In panel (b) this creates a zero-density interval, so the marker is the last registered boundary p_1^-; a unique simple zero is again registered after the first completed step. The accepted model has $\nu_1 = \nu_2 = 0$, and the gold curve is consequently zero in every panel. Translation-mode coupling to book 2 remains active and is retained in the numerical ledger but is not drawn as an additional curve. The registered boundary moves by 0.0997 after the first completed step, reaches 0.111 at 2 seconds and relaxes to a residual displacement of 0.0160 at 80 seconds. The price grid and operational step are fixed. All contribution curves use the same density scale and are not independently rescaled. The figure recovers the earlier panel structure without claiming a pointwise replication of the Bauer et al. trajectory.

9.2 Stylised facts under operational and calendar observation

Figure 13 retains the panel hierarchy of Bauer et al. as a structural reference while changing the estimands to match the current theory and available provenance. The earlier figure compares empirical data with a calibrated non-uniform-time legacy implementation. The present figure compares one current-model ensemble before and after a separate previous-refresh observation map. The operational model evolves on a fixed grid; calendar non-uniformity enters only through subordination. Log-mid displacement is reported relative to the translation-invariant zero origin, so its level cannot be compared with the arbitrary price levels in the legacy figure.

The production event process uses irregular background market-order arrivals at the declared mean rate. Event quantity is fixed from the full-experiment event/background displacement scale rather than from the earlier impact-test quantity. No smoothing, path selection or parameter refit is used. The accepted ensemble contains 16 paths and 5,600 operational steps per path; the extension from 2,800 steps was required to obtain at least 25 operational observations beyond $|z| > 3$ under the registered tail gate.

The distinction between the two rows explains the principal morphological difference. The operational row has no zero-return atom and is close to Gaussian for the registered baseline. The calendar row holds the last available value between refreshes, creating 60.49% zero five-second increments, a central density spike, an extended QQ plateau and larger standardised tails. These are observation-clock effects; they do not imply that the underlying operational path has changed.

The baseline uses ordinary operational order $\alpha_u = 1$, zero cancellation and a Poisson refresh clock. It therefore does not test fractional operational memory or non-Markovian calendar waiting times. Those mechanisms remain distinct in the Angstmann–Gebbie formulation and require their own registered numerical controls before use. Signed-order-flow persistence is imposed by the declared finite-persistence aggressor-sign input and is not an endogenous model result.

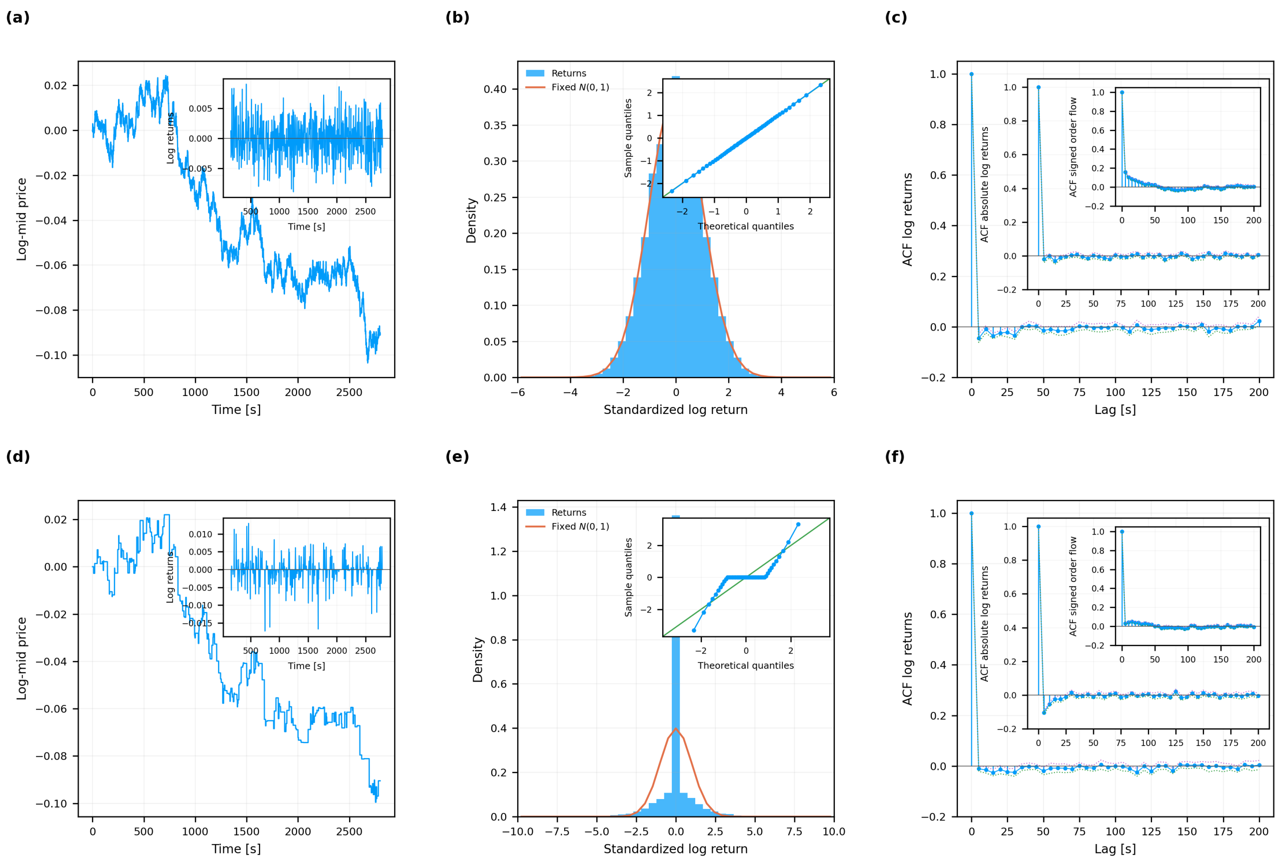

Figure 13: **Current-model stylised facts under operational and calendar observation.** Panels (a)–(c) use five-second increments on the uniform operational grid; panels (d)–(f) use previous-refresh calendar observations of the same simulated paths. Panels (a) and (d) show the Book 1 log-mid displacement for the registered representative path, with unstandardised log returns inset. Panels (b) and (e) show pooled standardised-return densities with the fixed standard-normal reference and normal-QQ insets. Panels (c) and (f) show ensemble log-return ACFs, with absolute-log-return and declared- aggressor signed-order-flow ACFs nested in the same order as Bauer et al. Blue curves are model estimates; orange is the standard-normal reference, green is the QQ identity or lower pointwise uncertainty curve, and magenta is the upper pointwise uncertainty curve. The fixed-grid implementation uses $\alpha_u = 1$ and zero cancellation. The calendar path is stepwise because previous-refresh observation holds the last operational log-mid value between asynchronous refreshes. This creates a central zero-return mass and stronger calendar-time tail curvature without changing the underlying operational path. Unlike Bauer et al. Figure 3, the rows are an operational/calendar comparison rather than empirical/calibrated data; no empirical row, legacy fractional recurrence or absolute price-level claim is made. Signed-order-flow persistence is imposed by the declared finite-persistence input and is not an endogenous result.

Table 1: **Parameter, timescale and identifiability register.** The table maps paper notation to implementation names and records units, roles, sensitivity pathways, and fixed or swept status. Refresh rate and source amplitude receive different code names despite source-notation overlap. Identifiability classifications describe the information available from the planned outputs under the stated model; they are not estimates, formal statistical-identifiability proofs, or empirical calibration results.

Paper symbol	Code name	Source	Units	Role	Sensitivity path	Figure treatment	Identifiability from Epps output
Δ	`aggregation_scale`	Eqs. (8), (9), (11)	time; dimensionless after scaling	return aggregation horizon	direct horizontal scale	swept in F1-F6	observed/design variable
ρ_∞	`rho_inf`	Eqs. (8), (9), (11)	dimensionless correlation	long-scale vertical level	multiplicative vertical scale	fixed to one in base figures	conditional on plateau coverage
λ_{12}	`clock_rate`	Eq. (8); App. A	time^{-1}	pooled refresh/overlap rate	$F(\lambda_{12}\Delta)$	unit scale; sensitivity in F2	conditional; independently measurable in v2
$\kappa = \kappa_1 + \kappa_2$	`response_rate`	Eq. (9); App. B	time^{-1}	spread relaxation rate	$F(\kappa\Delta)$	swept in F1-F2; derived in F4-F5	conditional on clock rate and short scales
α_c	`alpha_clock`	App. C, Eq. (C.18)	dimensionless	fractional clock order	short power Δ^{α_c}	swept in F3	not separate from alpha_response at short scale
α_r	`alpha_response`	App. C, Eq. (C.18)	dimensionless	fractional response order	short power Δ^{α_r}	swept in F3	not separate from alpha_clock at short scale
τ_c, τ_r	`clock_characteristic_time` `response_characteristic_time`	v1 nondimensionalisation	time	fractional scale convention	$(\Delta/\tau)^{\alpha}$	fixed to one in F3	confounded with fractional rate convention
γ_{jk}	`coupling_strength`	Eq. (6); App. B	model-dependent	pair-trader coupling strength	$\kappa_j \propto \gamma_{jk}$	one-at-a-time sweep in F4	not separate from source/front quantities
λ_j (source)	`source_amplitude`	Eq. (5); App. B	source amplitude	decaying-Gaussian amplitude	$M_j^+ \propto a_j$	one-at-a-time sweep in F4	not separate from coupling strength
μ_j	`source_width`	Eq. (5); App. B	price^{-2}	Gaussian source-shape scale	$M_j^+ \propto \mu_j^{-1/2}$	one-at-a-time sweep in F4	not separate from amplitude/front slope
$\lvert\mathcal{L}_j\rvert$	`front_slope_abs`	Eq. (B.9); Eq. (B.13)	density per price	frozen reaction-front slope	$\kappa_j \propto \lvert\mathcal{L}_j\rvert^{-1}$	one-at-a-time sweep in F4	not identifiable as thickness from Epps alone
ε	`epsilon`	Eq. (6); App. D	price^2 for yz/ε	continuum selector regularisation	centred moment invariant; representation path	swept in F5	not a reaction-front thickness parameter
Δx	`lattice_spacing`	App. D	price	numerical lattice resolution	discrete moment error to effective response	swept in F5	known numerical control
z_{jk}	`directed_spread`	Eq. (6); App. B	price	directed book-price spread	selector argument yz/ε	fixed to 0.2 in F5	state variable, not inferred parameter
–	`alignment_offset_cells`	v1 numerical diagnostic	lattice cells	source-selector centring error	parity breaking to effective response	0 and 0.5 in F5	known numerical control
–	`domain_halfwidth`	v1 numerical diagnostic	price	finite integration/lattice domain	tail truncation to effective response	full and truncated in F5	known numerical control

Table 2: **Numerical benchmarks, convergence tests and acceptance status.** Each row records a claim-relevant diagnostic, its analytic or independent reference result, declared tolerance, observed value or error, and acceptance status. Checks cover the ordinary kernel and derivative, fractional $\alpha = 1$ recovery and independent reference values, the decaying-Gaussian first moment, centred-selector invariance, finite-grid refinement, invalid inputs, and the analytic/simulation overlay contract. The table supports numerical reliability and reproducible execution within the tested environment; it is not a mathematical proof of the model, an empirical validation, or evidence for excluded legacy-simulation claims.

ID	Diagnostic	Expected result	Tolerance	Observed result	Status
D01	Ordinary zero and small-argument limit	F(0)=0 and the declared small-x series is recovered	abs error <= 1.0e-12	F(0)=0.000e+00; series error at 1e-6=0.000e+00	Verified
D02	Ordinary bounds and monotonicity	0 <= F(x) < 1 and non-decreasing for x >= 0	violation <= 1.0e-12	range=[0.00000500, 0.99000000]; min increment=1.641e-07	Verified
D03	Ordinary analytic derivative	Analytic derivative agrees with centred finite differences	max error <= 2.0e-07	max error=2.084e-10	Verified
D04	Ordinary elasticity limits	S_F -> 1 at short scale and S_F -> 0 at long scale	combined limit error <= 2.0e-05	S(1e-8)=0.999999996667; S(1e5)=1.000e-05	Verified
D05	Fractional alpha-one recovery	F_alpha with alpha=1 equals the ordinary kernel	max error <= 2.0e-13	max error=0.000e+00	Verified
D06	Fractional independent reference values	Release evaluator matches adaptive-quadrature references	max error <= 2.0e-09	six-point max error=1.571e-13	Verified
D07	Fractional bounds and monotonicity	Release curves remain in [0,1] and non-decreasing	violation <= 2.0e-09	maximum violation=0.000e+00	Verified
D08	Decaying-Gaussian first moment	Numerical full-domain selected moment equals analytic half-line moment	relative error <= 2.0e-06	analytic=-0.443113462726; numerical=-0.443113462726	Verified
D09	Centred-selector continuum invariance	Symmetric first moment is independent of selector width	relative error <= 2.0e-06	max relative error over three epsilon values=2.220e-16	Verified
D10	Centred finite-grid convergence	Moment error decreases from a deliberately coarse lattice under refinement	finest error <= 2.0e-05	errors=1.935e-01,1.939e-03,1.635e-06,1.110e-15	Verified
D11	Boundary baseline response normalisation	Two-book baseline total response rate equals one	abs error <= 1.0e-12	kappa_total=1.000000000000	Verified
D12	Combined-factor construction	Combined curve equals diagnosed component product and remains bounded	violation <= 1.0e-12	range=[0.00000025, 0.98010000]	Verified
D13	Invalid-input handling	Four invalid parameter cases fail clearly	all four raise ValueError	caught=4/4	Verified
D14	v2 analytic/simulation overlay contract	Required scientific and provenance roles are declared	all ten roles present	declared roles=10/10	Verified

10 Parameter and numerical evidence

Table 1 keeps the source notation, unambiguous code names, units, sensitivity paths, and identifiability qualifications together. In particular, the paper's refresh-rate and source-amplitude uses of λ are separated in code. Table 2 is generated from the diagnostic results; scientific figures are not accepted if a row is failed. The fractional reference points were computed independently with adaptive quadrature, while the release evaluator uses fixed composite Gauss–Legendre quadrature and a small-argument series.

11 Reproducibility contract

The complete active route is

```
python scripts/run_all.py
```

and the standard-library regression tests are

```
python -m unittest discover -s tests -v
```

The run regenerates or scientifically verifies the thirteen selected PDF/PNG figure pairs, two CSV/LaTeX table pairs, claim-bearing curve and summary datasets, diagnostic reports, and the retained simulation route. PDF bytes can differ across systems because of renderer metadata or font embedding; frozen inputs, machine-readable values, declared tolerances and diagnostic statuses are the cross-platform reproducibility objects.

The accepted analytical output schemas retain their scientific object versions. The final simulation evidence is registered in the component, combined, impact and dependence CSV files listed in the figure manifest. Simulation resolution, alignment and domain metadata accompany the comparisons because Figure 5 shows that numerical boundary representation can move the apparent response curve.

12 Interpretive limits

The diagnosed computations support a limited set of conclusions. Ordinary clock and coupling factors are distinguishable when their timescales are sufficiently separated and the components are shown explicitly. The long-scale plateau contains little local information about either rate. The leading short-scale power of a combined fractional curve identifies an order sum rather than two separate orders. Reaction-boundary quantities affect the Epps curve conditionally through the effective response rate, but are not individually identified by that curve. Continuum selector invariance can coexist with finite-grid distortion when parity, alignment or domain controls are not respected. The translation-mode simulation is consistent with the registered component curves and improves the no-refit combined comparison when the estimator's finite-grid, finite-step clock selection is represented exactly. The impact, shock-recovery, autocorrelation and stylised-facts figures are conditional model diagnostics, not empirical calibration.

These are computational illustrations and numerical verification statements inside the paper's assumptions. They are not empirical validation, calibration or proof of the approximations. The retired Bauer implementation is not required to execute or interpret the accepted v2 route.